

TikTok Claims Classification | Job Proposal

Hello,

I can help build a practical machine learning workflow to classify user-reported content into meaningful categories and turn the analysis into a model that can support review prioritization.

For a similar classification case study, I worked through the full pipeline rather than stopping at model training: data-quality assessment, exploratory analysis, statistical testing, NLP feature extraction, structured feature preparation, model comparison, validation-based model selection, held-out test evaluation, and feature-importance analysis.

The workflow used CountVectorizer to convert transcription text into 4,014 training-derived features and compared Random Forest with XGBoost. Model selection was performed on validation data, and the final model was evaluated once on an untouched test set. The selected Random Forest achieved 99.95% accuracy, 100% precision, 99.90% recall, and 99.95% F1 on that test set.

For a client project, I would focus on making the result useful beyond the metric itself:

- define the target and prediction point clearly
- inspect data quality and potential leakage
- engineer text and structured features appropriate to the task
- compare models using metrics aligned with the business objective
- evaluate the final model on unseen data
- explain the strongest predictive signals
- document limitations and deployment considerations
- provide a reproducible notebook and model artifacts

I would also verify whether each feature is actually available at prediction time before recommending deployment. High offline performance is useful only when the same information can be obtained reliably in the real workflow.

Deliverables can include a cleaned analysis notebook, trained model, evaluation report, feature-importance analysis, and a concise business summary.

Best,

Mounir

Data Science | Machine Learning | Advanced Data Analytics